Comparison of Static Routing and Dynamic Routing

Feature	Static Routing	Dynamic Routing (RIP/OSPF)
Definition	Routes are manually configured by the network administrator.	Routes are learned and updated automatically using routing protocols like RIP or OSPF.
Configuration	Manual	Automatic
Best For	Small and stable networks	Large and changing networks
Route Updates	Manual updates are required	Updates happen automatically
CPU & Bandwidth Usage	Low	Higher due to routing updates
Scalability	Poor for large networks	Excellent for large networks
Failure Recovery	Manual intervention required	Automatically finds an alternate path
Security	More secure because routes are fixed	Slightly less secure due to routing advertisements

Static Routing	**Dynamic Routing (RIP/OSPF)**
Pros	**Pros**
✔ Easy to configure for small networks	✔ Automatically updates routes when the network changes
✔ Uses less CPU and bandwidth	✔ Ideal for large and complex networks
✔ More secure because routes are manually controlled	✔ Automatically reroutes traffic if a link fails
Cons	**Cons**
✗ Requires manual updates whenever the network changes	✗ More complex to configure
✗ Not suitable for large networks	✗ Uses more CPU and bandwidth
✗ No automatic failover if a link fails	✗ May take time to converge after network changes

Application	Static Routing	Dynamic Routing
Food Delivery App (Zomato/Swiggy)	If the connection between city servers fails, an administrator must manually update routes before orders can continue.	If a connection fails, routers automatically choose another path, allowing orders to continue with minimal interruption.
Small Office Network	Suitable because the network rarely changes.	Usually unnecessary due to the small network size.
Large Company (Google/Amazon)	Difficult to manage because of many routers and networks.	Preferred because routes are updated automatically across the entire network.